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COMP4139

Machine Learning (MLE)

Challenges & Trends in Machine Learning

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Challenges



Outline of Challenges

Many application scenarios have limited data (e.g. healthcare)

Annotation is often costly, time consuming and requires expertise

Model can be generalised in practice

Research institutions have no thousands of GPUs to train large models.

AI is useful, but human is still needed, right?

Do you trust AI? What if it goes wrong?

Model Training Using Small Dataset

Many application scenarios have limited data (e.g. healthcare)

- Data augmentation and synthesis.
 - Geometric, intensity, add noise, etc.
 - Generative models learned from a similar task or start from a small dataset.
- Transfer learning, big foundation models.
 - Use pretrained models based on large datasets and fine-tuned on small dataset
 - E.g. Segmentation Anything Model, ChatGPT, Models trained on ImageNet, etc.
- Small annotated data but large unannotated data
 - Semi-supervised learning: pseudo labelling, teacher-student model.
 - Zero-shot, one-shot, few-shot learning.

Reduce Data Annotation Efforts

Annotation is often costly, time consuming and requires expertise

- Inaccurate annotation
 - Limited by expertise, time, and cost.
 - Challenging by the nature of the task.
- Weakly supervised learning
 - Only provide image level label to achieve detailed object segmentation.
 - Only provide a rough annotation (bounding box, scribble, etc) to achieve detailed object segmentation.
- Active Learning and Representative Sample Selection
 - Use clustering method to select representative examples for annotation.



Model Generalisability

Model can be generalised in practice without frequent model retraining

- Distribution shift between training and testing
 - Limited data for training
- Utilise pretrained models + model fine tuning
 - LLM, CNNs trained on ImageNet, Stable diffusion models, SAM
 - Fine tune which part of the model is tricky: all parameters, all decoder layer, final layer, etc.
 - Domain adaptation
 - Cross domain data synthesis (e.g. cycleGAN, conditional diffusion models, etc.)
 - Adding noise, corrupt the data (e.g. Masked Autoencoder)
 - Missing information in practice
 - Missing image modality, certain measures, follow-up data
 - Apply data imputation, synthesis, ensemble learning, etc.



Limited Computing Resources (Training/Deployment)

Not all research institutions have hundreds/thousands of GPUs to train large models. It has been dominant by large companies.

- We could fine-tune these large models but still hard.
- How much data we need? Which part of the model we fine-tune?

Cloud computing is commonly used for model deployment, but in many applications the model needs to run on low cost and embedded devices. E.g. autonomous driving.

- Model distillation and model pruning to simplify the model
- Remove redundant information/features

AI is useful, but human is still needed, right?

- AI as a generation tool based on human's instructions
 - ChatGPT to draft email, letters, ppt, paper, code, then human to refine it.
 - Painting, art generation based on creative prompt from human.
- AI is treated as another worker
 - AI has been tested in breast cancer screening and it outperforms an average radiographer.
- AI as a pre-processing tool to reduce human's workload
 - AI handles a large number of easy cases and leave fewer challenging cases for human.



Do you trust AI, including generated content, self-driving, robot? What if it goes wrong?

- Most AI models are not self-aware of its own limitations.
 - For every input, it will produce an output.
 - The output likelihood value is not equivalent to certainty and reliability.
- Aware of the distribution of the training data.
 - If the new input is out of the distribution, flag it up.
- Uncertainty estimation
 - Aleatoric uncertainty due to randomness. (introduce model randomness helps)
 - Epistemic uncertainty refers to lack of knowledge. (more data helps)



“typewriter keyboard”
with certainty 83.14%.



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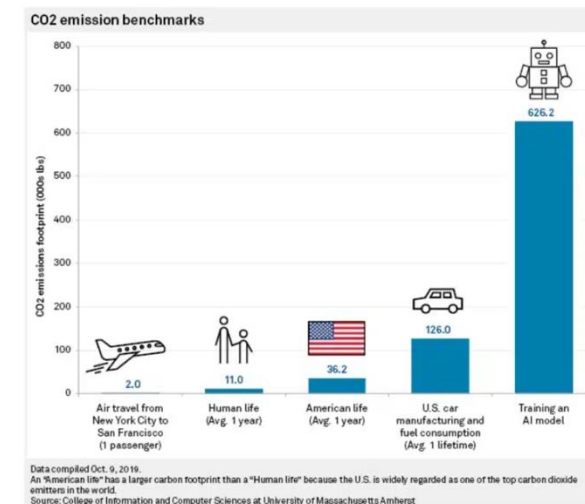
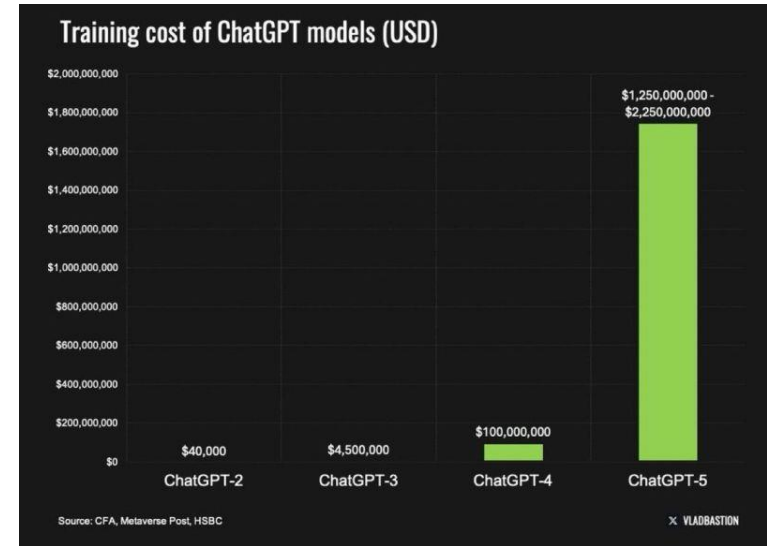
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Trends



Generative AI- Requirement for Model Training

- Extremely large Data
 - ChatGPT used **terabytes** of data, including web, books, etc.
 - DALL-E 2 used ~650 million image-text pairs to train.
 - Sora used >32 million videos plus thousands of video game recordings.
- A lot of GPUs and energy for training
 - ChatGPT3 has >175 billion parameters, requires 30,000 Nvidia A100 (\$10K each)
- Manual labour to rectify/regulate the generated content.



ChatGPT: Instruction-tuned decoder LLMs using fine-tuning + RL

GPT-4 Large decoder transformer family [GPT-4 Technical Report \(OpenAI\)](#)

LLaMA (Meta) : Transformer family (7B–65B etc.) [LLaMA paper - \(arXiv\)](#)

Gemini (Google / DeepMind): Family of multimodal transformer models [Gemini paper - arXiv](#)

Claude 3 (Anthropic): Transformer family of large multimodal models. [Claude 3 \(Anthropic\)](#).

DeepSeek V3 Mixture-of-Experts transformer designs [DeepSeek technical reports / Hugging Face repo \(arXiv\)](#)

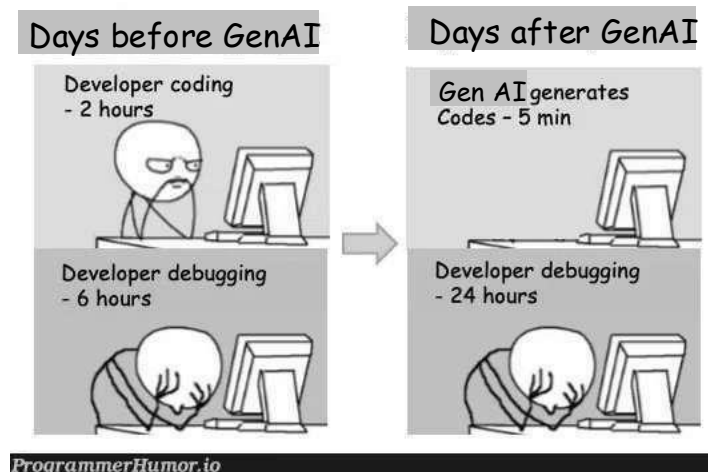
and so on...

GitHub Copilot (Codex → Copilot) Product built on code-specialised LLMs [initial work –arxiv](#)

Code Llama (Meta) LLaMA-based decoder transformer fine-tuned on code [Code Llama – arxiv](#)

Cursor / Windsurf etc. Editoric AI products built on LLMs with retrieval-augmented workflows and agentic tool chains

Gemini Antigravity - last week -(code variants) Gemini family includes code-specialised variants



Stable Diffusion Latent Diffusion Model (VAE encoder/decoder + U-Net denoiser operating in latent space; cross-attention for conditioning). [Latent Diffusion paper - CVPR \(arXiv\)](#)

Imagen (Google) Text→image diffusion model combining large text encoder (T5) with diffusion image generator (cascaded diffusion approach) [Imagen paper - \(arXiv\)](#)

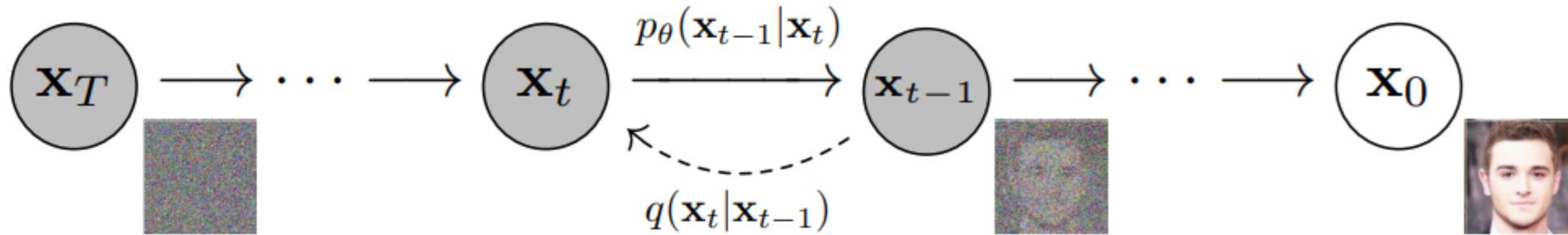
DALL·E 2 (OpenAI) Two-stage approach in OpenAI's DALL·E-2: a prior that maps text→CLIP embedding, and a decoder (GLIDE-based diffusion) conditioned on that embedding [DALL·E-2 paper - \(arXiv\)](#)

DALL·E 3 (OpenAI) Productised multimodal transformer / direct generation approach [DALL·E 3 \(arXiv\)](#)

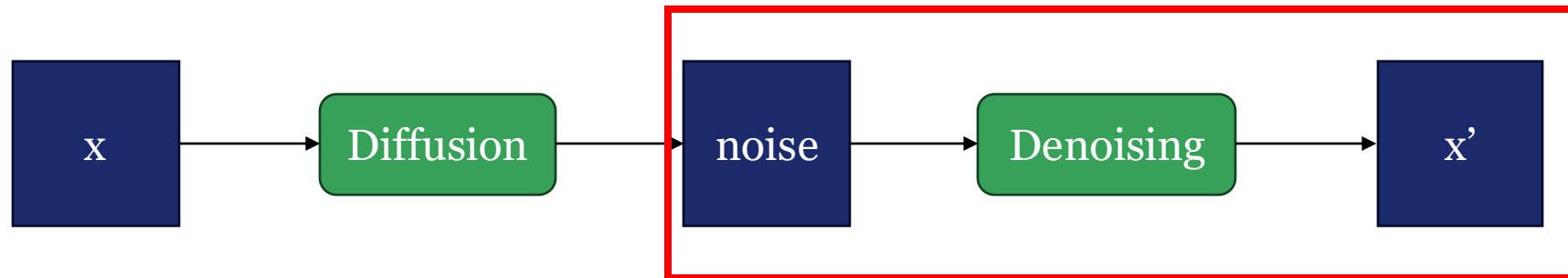
Sora (OpenAI) <https://openai.com/index/sora/>

Lumiere (Google Research) Space-Time U-Net diffusion model that generates the full temporal duration in a single space-time pass [Lumiere: Space-Time \(arXiv\)](#)

Aim: Noise to Image generation



Inference



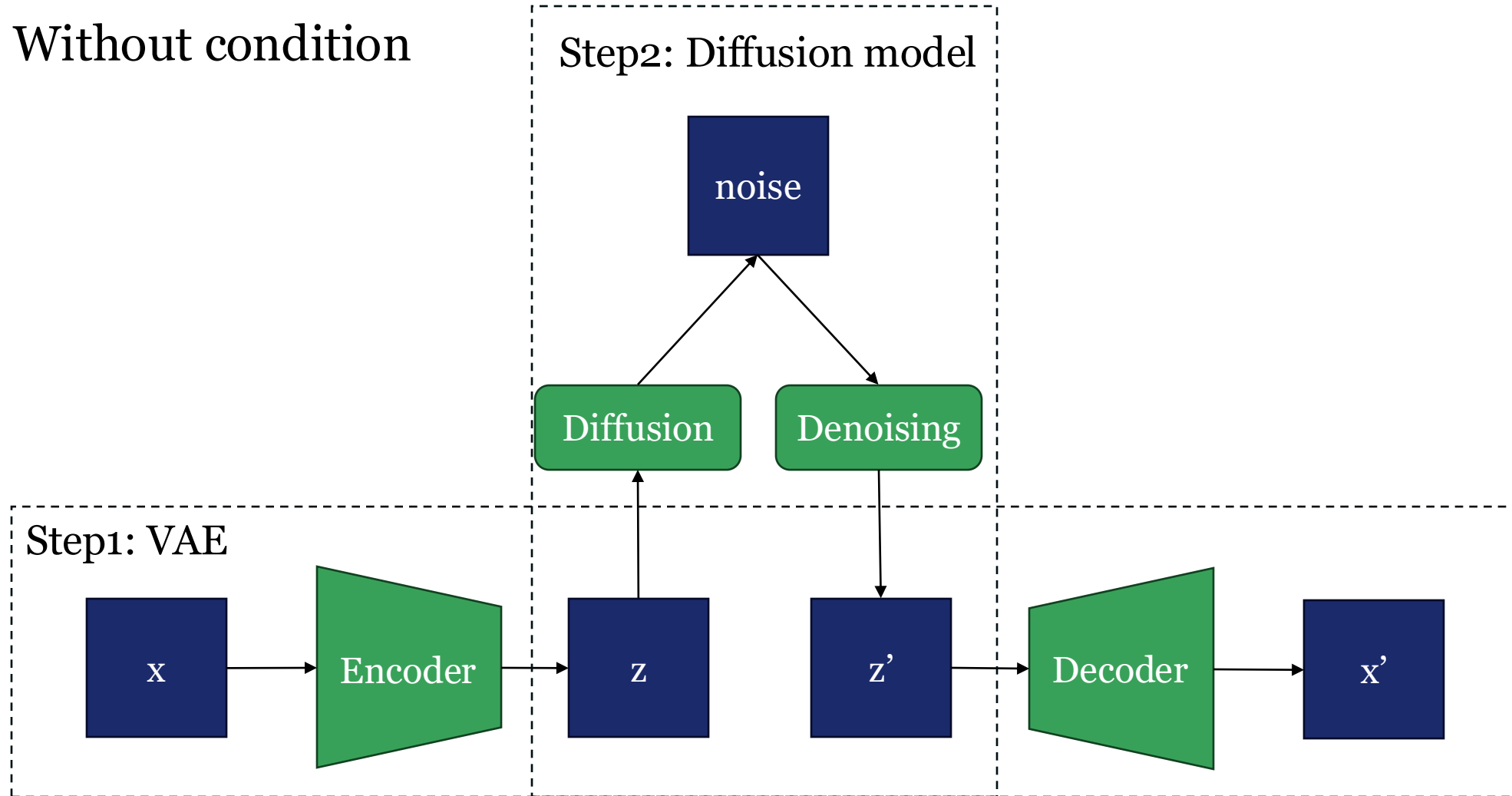
Drawback:

- Lack details for high-resolution images.
- Computationally expensive.



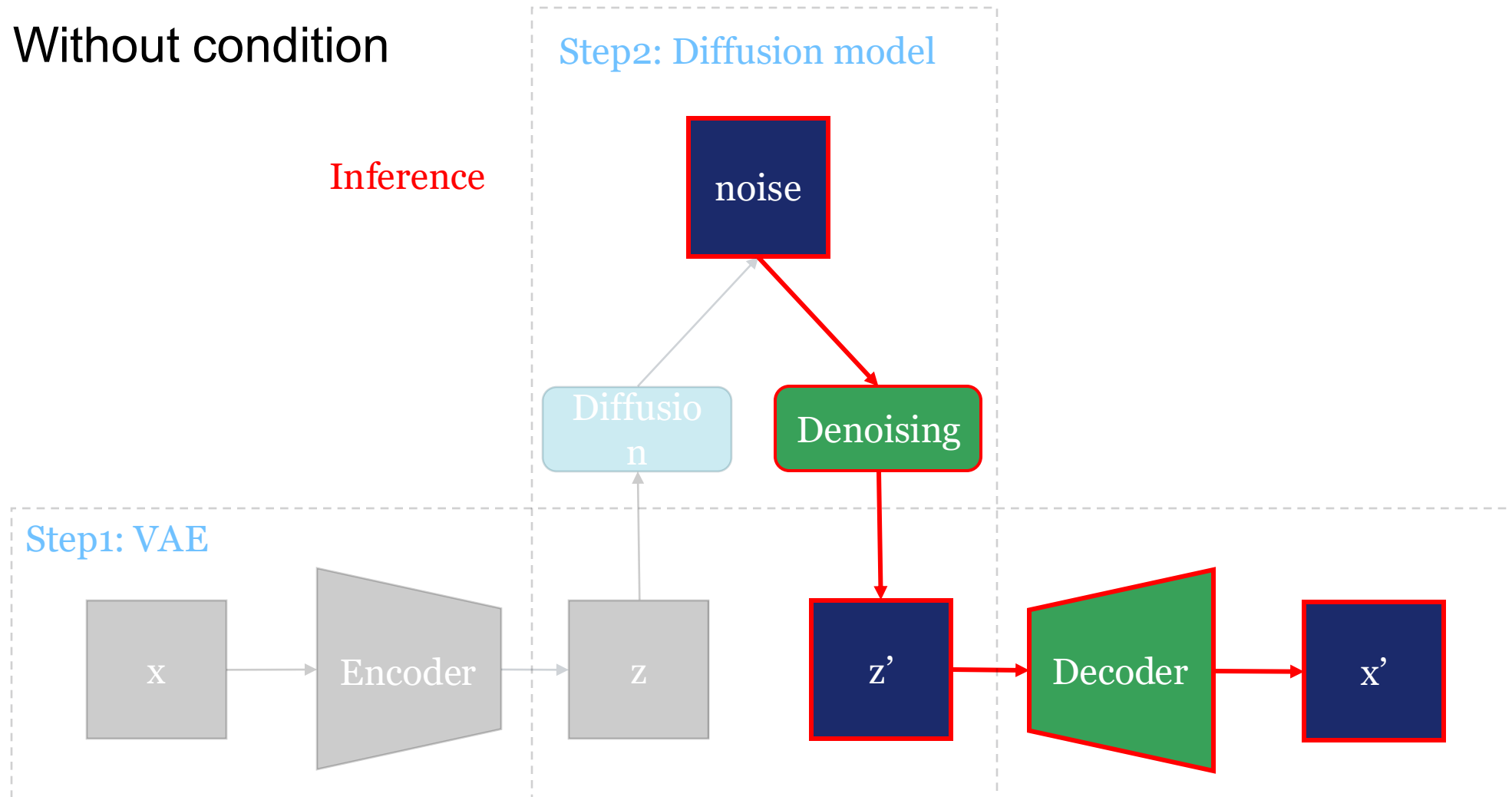
Stable Diffusion

- Without condition

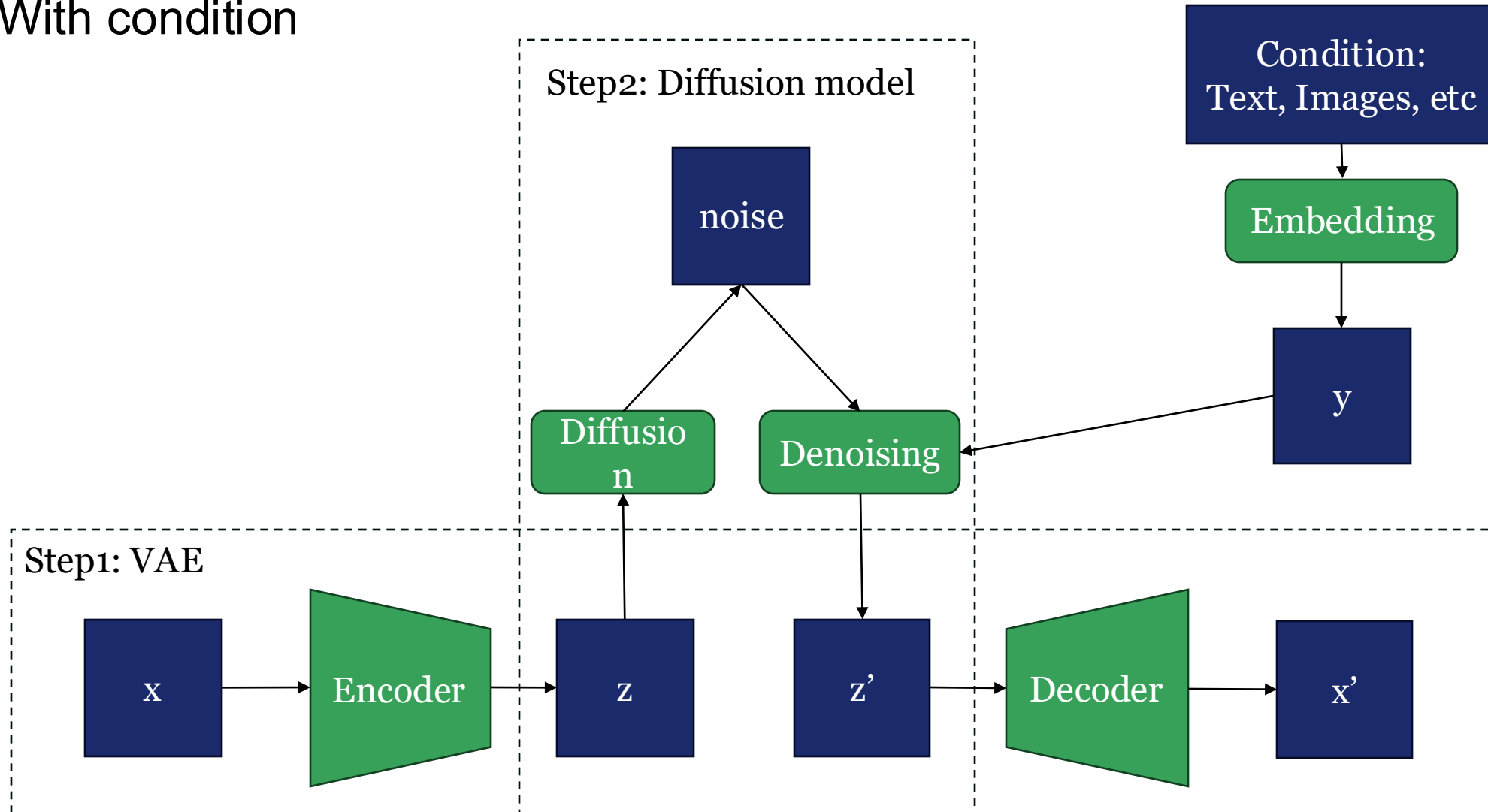


Stable Diffusion

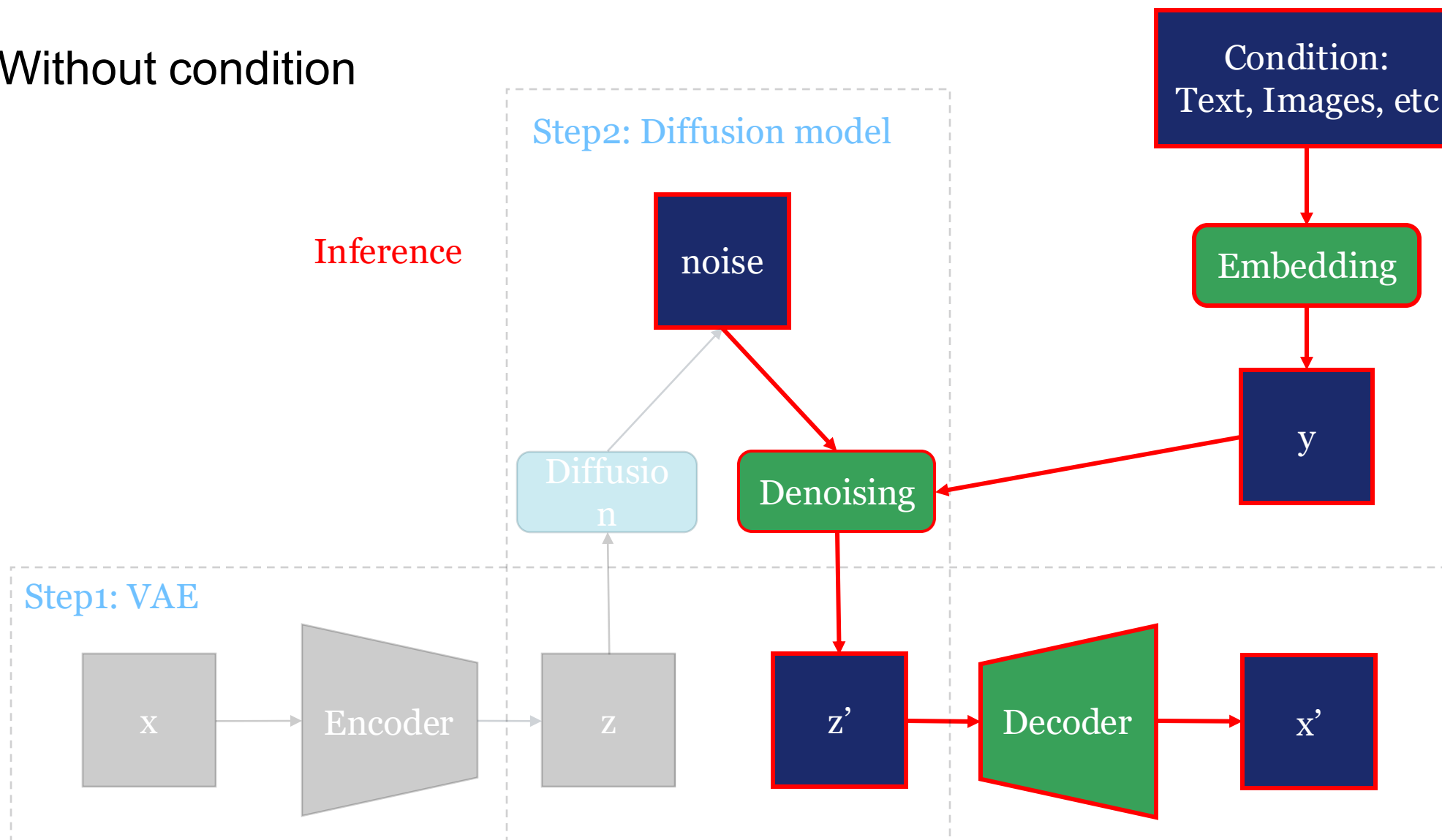
- Without condition



- With condition

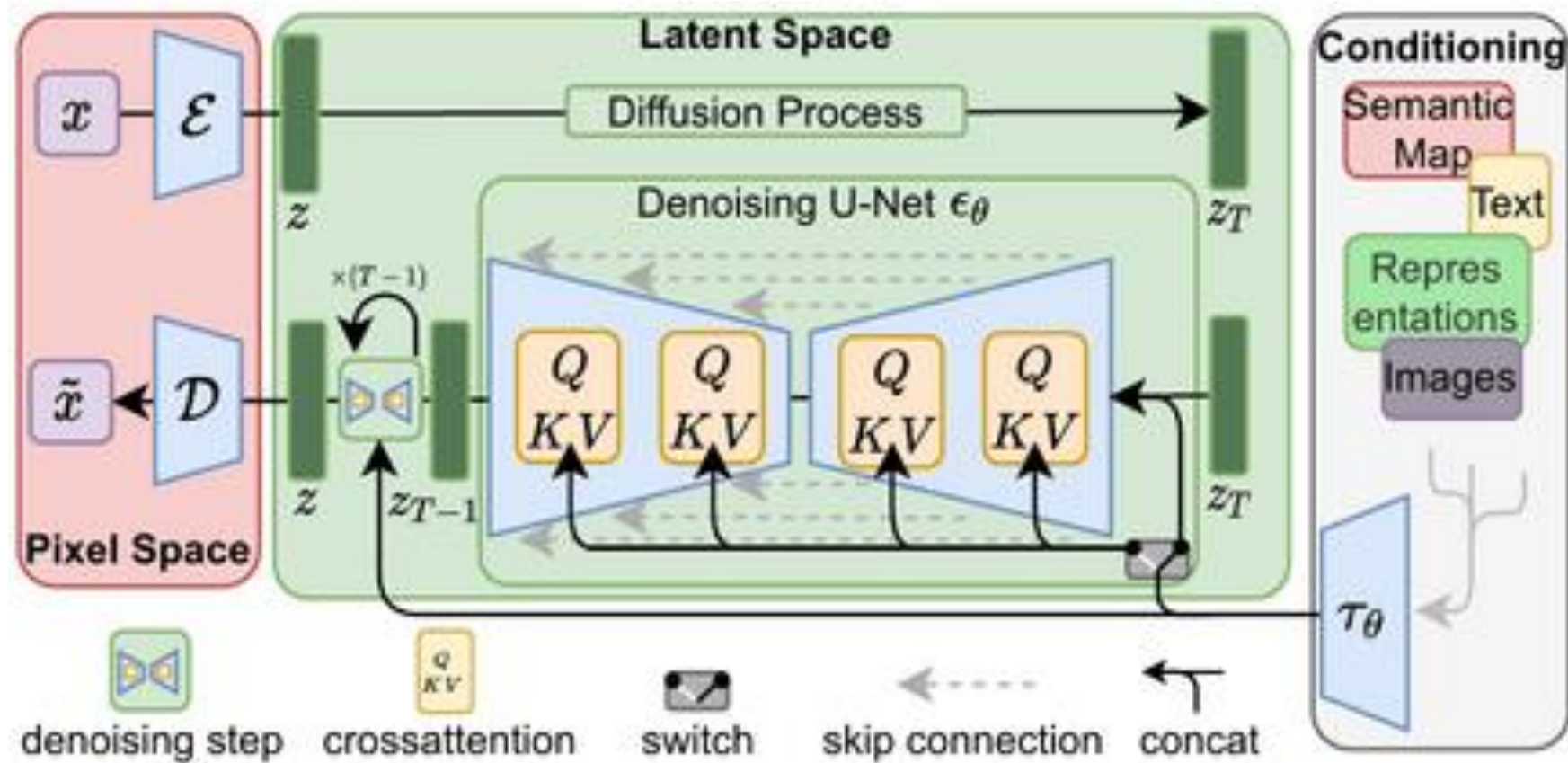


- Without condition



Stable Diffusion Model (Latent Diffusion)

- Stable diffusion enables conditioning: <https://stability.ai/>



Stable diffusion: VAE+UNet



- Video generation models as world simulators
 - Generate one minute video; other models produce content around 5 seconds.
 - Simulate the physical world using game engine.
- Training data
 - Video + frame level description (text) + summary (text).
 - Video: High resolution video (1080p)
 - Description: using caption for each frame from DALL-E 3 (re-captioning technique).
 - Using GPT extend the prompts.

- Video to patch in both spatial and temporal directions
- Then patches to VAE + Diffusion Transform blocks

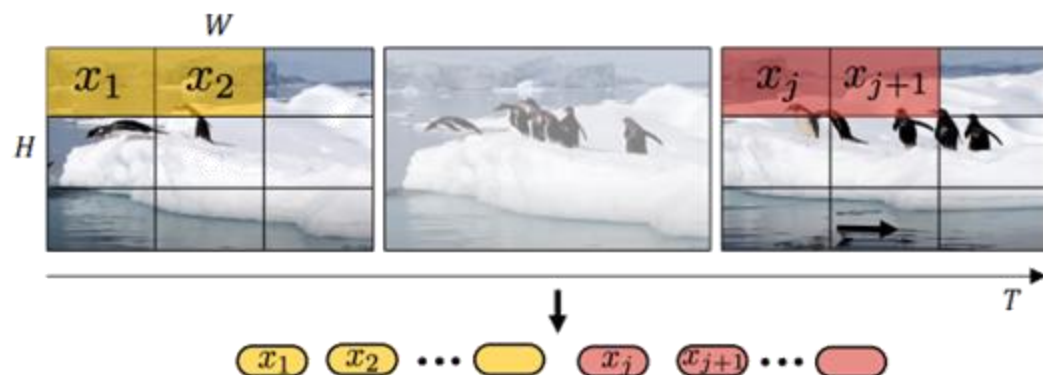


Figure 2: Uniform frame sampling: We simply sample n_t frames, and embed each 2D frame independently following ViT [18].

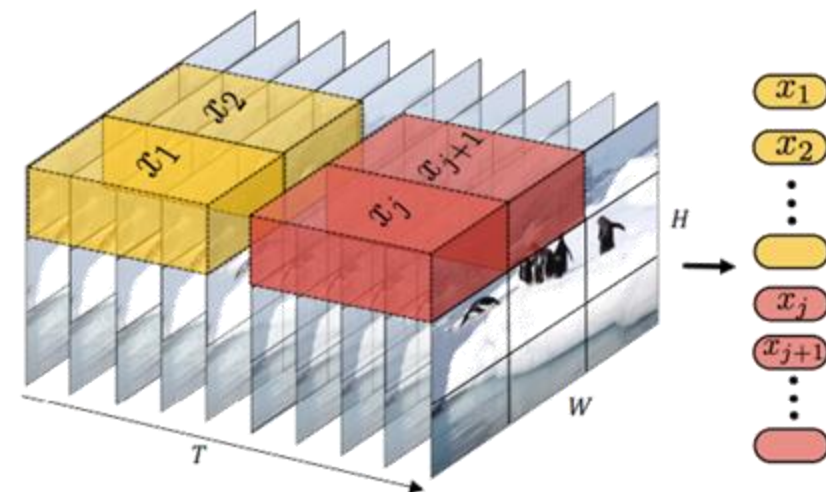


Figure 3: Tubelet embedding. We extract and linearly embed non-overlapping tubelets that span the spatio-temporal input volume.



- Usage:
 - Generate video using prompt.
 - Animating DALL·E images.
 - Extending generated videos.
 - Video-to-video editing.
 - Connecting videos.
- Limitations:
 - Does not accurately model the physics of many basic interactions
 - Do not always yield correct changes in object state.
 - Incoherencies that develop in long duration samples or spontaneous appearances of objects.



Multimodal Data Integration

- ChatGPTo combines visual, audio and text
 - Other data types: sensors, knowledge base, physical forces, etc.
- Multimodal data embedding and fusion
 - Feature representation and embedding
 - Feature concatenation or cross attention: e.g. CLIP
- Handling missing modality
 - Inconsistency between training and testing
 - Often lack of complete data in practice



AGI



Artificial General Intelligence (from Amazon AWS)

- No consensus definition, something like human-like intelligence and the ability to self-teach.
- Not trained for specific task using a specific dataset.
- Approaches in AGI research
 - Symbolic (logic network symbolizes physical objects with an if-else logic)
 - Connectionist (replicate human brain with neural networks like ChatGPT)
 - Universalists (theoretical calculation of AGI complexities, Einstein style)
 - Physical integration (robotics to allow physical interactions)
- Challenges
 - Generalisability across domains, like human.
 - Sensor technology to allow full physical interaction.
 - Emotional intelligence is the next level.

The Illusion of Thinking



What if we achieved AGI?

February 24, 2023

Planning for AGI and beyond

Our mission is to ensure that artificial general intelligence—AI systems that are generally smarter than humans—benefits all of humanity.

<https://openai.com/index/planning-for-agi-and-beyond/>

- https://www.ted.com/talks/fei_fei_li_with_spatial_intelligence_ai_will_understand_the_real_world?s_subtitle=en&trigger=30s
- <https://www.youtube.com/watch?v=Y6Sgp7y178k>
- <https://www.youtube.com/watch?v=ketW8xsL-ig>



Learning Objectives

No official learning objectives, but many points to explore